Question	558
Type	MCQ
Body	The value of $\int_{-5}^{5}\left(t^3+6t^2+2t+\cos\pi t\right)b''\left(t-2\right)dt$ is
Option	0
Option	$24-\pi^2$
Option	24
Option	$12-\pi^2$
Correct	2
Explanation	$\int_{-\infty}^{\infty}x\left(t\right)\delta''\left(t-t_0\right)dt=(-1)^2x''\left(t_0\right)$ Let $x\left(t\right)=t^3+6t^2+2t+\cos\pi t$ $x'\left(t\right)=3t^2+12t+2-\pi\sin\pi t$ $x''\left(t\right)=6t+12-\pi^2\cos\pi t$ $x''\left(2\right)=24-\pi^2$ $\int_{-5}^{5}\left(t^3+6t^2+2t+\cos\pi t\right)\delta''\left(t-2\right)dt=24-\pi^2$
Subject	Signals and Systems
Topic	
Correct Marks	2
Incorrect Marks	0.66
Hint	-
Video Solution	-
PYQ	
OTS ID	